Impact of urban environments on Vehicle-to-Vehicle (V2V) communication

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***Abstract*—Vehicle-to-Vehicle (V2V) communication is vital for autonomous safety, yet highly susceptible to urban environmental physics. This paper analyzes IEEE 802.11p DSRC signal degradation across three path-loss scenarios (Free Space, Two-Ray Ground, and Obstacle Shadowing) using the Veins simulation framework. While idealized free-space models achieve near-perfect packet delivery, introducing realistic asphalt reflections degrades signal quality through destructive interference. Furthermore, building blockages physically sever line-of-sight connections, dropping successful communication by roughly 67%. These findings prove the absolute necessity of employing realistic physical-layer analogue models when evaluating VANET protocols.**

***Index Terms*—V2V, VANET, IEEE 802.11p, DSRC, OMNeT++, Veins, Obstacle Shadowing, Multipath Fading**

I. Introduction

Cooperative Intelligent Transport Systems (C-ITS) rely heavily on the real-time exchange of safety beacons, such as Basic Safety Messages (BSMs), to prevent collisions and manage traffic flow. This communication is typically handled by the IEEE 802.11p standard operating in the 5.9 GHz frequency band.

Because 5.9 GHz radio waves require a near line-of-sight (LOS) to communicate effectively, the physical environment of a city becomes the greatest bottleneck to network reliability. Many early network simulations assumed vehicles communicate in a ”vacuum” (Free Space), leading to overly optimistic results that fail in real-world deployment.

In this study, we transition from theoretical, idealized network models to highly realistic physical environments. We systematically isolate and measure the exact impact of asphalt reflections and concrete buildings on MAC-layer packet drop rates. By proving how the physical environment actively destroys data, we demonstrate the limits of standard V2V communication in dense urban grids.

II. Related Work

The performance of Vehicular Ad-Hoc Networks (VANETs) has been the subject of extensive research, particularly regarding the limitations of the physical layer. The foundational guidelines for V2V communication are established by the IEEE 1609 family of standards and the 802.11p amendment [1], which dictate the Medium Access Control (MAC) and Physical (PHY) layer behaviors for Dedicated Short-Range Communications (DSRC).

Building upon this, further research by Sommer et al. [2] introduced the computationally efficient ”Obstacle Shadowing Model” for OMNeT++, which we directly utilize in this study. This model departs from overly optimistic free-space assumptions and accurately calculates radio wave attenuation based on the specific intersections of signals with simulated concrete building polygons. By utilizing these established frameworks, our study directly builds upon prior efforts to bring realistic multipath fading and physical signal absorption into VANET evaluation.

III. Problem Definition

To accurately model the physical network, we utilized a bi-directionally coupled simulator: SUMO (Simulation of Urban MObility) to handle realistic vehicle traffic physics, and OMNeT++ to handle the radio network stack. We assume that all vehicles are broadcasting standard safety beacons at regular intervals.

To isolate the environmental impact, we tested three specific analogue models at the Physical Layer (PHY):

1) **Free Space:** Assumes an empty void where signal strength decays purely over distance ($1/d^2$).
2) **Two-Ray Ground:** Introduces the asphalt street. The receiving antenna calculates the direct radio wave *plus* the radio wave bouncing off the ground, causing violent signal fluctuations (multipath fading).
3) **Obstacle Shadowing:** Introduces physical polygons (buildings) into the simulation. If a line-of-sight intersects a building wall, the signal is aggressively attenuated (absorbed).

A. Simulation Parameters

The exact configuration used for the OMNeT++ simulation is detailed below:

- **Simulation Time:** 100 seconds
- **Playground Size:** 600m x 600m
- **MAC Protocol:** IEEE 1609.4 / 802.11p
- **Transmitter Power:** 20 mW
- **Bitrate:** 6 Mbps
- **Frequency:** 5.89 GHz (DSRC band)
- **Mobility Model:** Veins (TraCI coupled with SUMO)

IV. Results

To evaluate the reliability of the network, we extracted the total successfully received messages (`receivedWSMs`) and compared them against the total packets destroyed by the physical environment (`TotalLostPackets`) at the MAC and PHY layers.

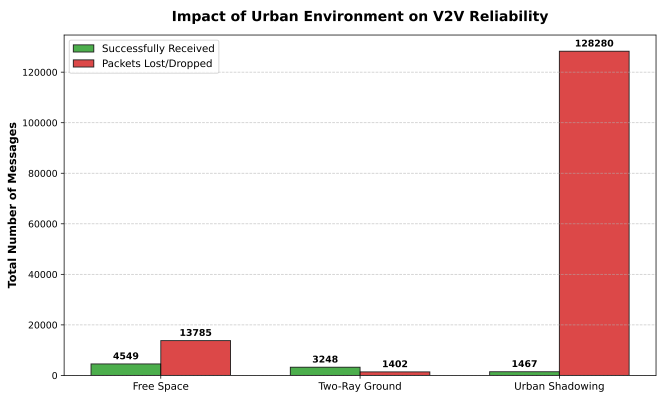

Fig. 1. Total successful message deliveries versus physically dropped packets across three path-loss environments.

A. The Free Space Baseline

As shown in Fig. 1, the Free Space model exhibits near-perfect communication. Because there is no interference or physical blockage, the vehicles successfully exchanged 4,549 messages. Packet loss was strictly limited to standard MAC-layer collisions (occurring when two cars accidentally broadcasted at the exact same millisecond).

B. The Impact of Ground Interference

When the Two-Ray Ground model is introduced, communication reliability visibly drops. The successful messages fell to 3,248. Because radio waves bounce off the asphalt, the receiving car's antenna experiences destructive interference. This lowers the Signal-to-Noise plus Interference Ratio (SNIR), causing the physical layer to corrupt and drop the packets before they reach the application layer.

C. The Urban Wall (Obstacle Shadowing)

The introduction of urban buildings resulted in catastrophic network degradation. Successful deliveries plummeted to 1,467 messages, accompanied by a massive spike in lost packets. Because 5.9 GHz signals cannot easily penetrate concrete, buildings physically severed the communication links between vehicles approaching intersections. This proves that distance is not the primary cause of packet loss in a city; line-of-sight blockage is.

V. Conclusion

This study evaluated the impact of environmental physics on IEEE 802.11p V2V networks. Through coupled simulation using SUMO and OMNeT++, we proved that network reliability is intrinsically tied to the physical surroundings. While idealized Free Space models showed excellent packet delivery, the introduction of realistic ground reflections reduced successful communication by roughly 28%. Furthermore, the addition of urban buildings caused the network to collapse by nearly 67% compared to the baseline, as line-of-sight blockages destroyed thousands of safety messages.

Ultimately, this demonstrates that V2V protocols cannot be designed in a vacuum. Future urban VANET designs must account for severe obstacle shadowing, potentially by relying on edge infrastructure like Roadside Units (RSUs) placed at intersections to relay messages around blind corners.

References

[1] IEEE Standard for Information technology–Telecommunications and information exchange between systems–Local and metropolitan area networks–Specific requirements–Part 11: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications Amendment 6: Wireless Access in Vehicular Environments, IEEE Std 802.11p-2010, 2010.

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